



Ensemble-Based Deepfake Detection Using Multi-Backbone CNNs & Meta-Learning

M.Tech Mini Project in Information Security & Privacy

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deepfake

Motivation



Deepfake Proliferation

Easy generation with open-source tools



Model Limitations

Single-model detectors lose 30pp accuracy on new data



Real-world Challenges

Videos undergo compression, blur, color jitter



Ensemble Advantage

Mitigates individual bias patterns

Project Objectives



Fine-tune CNNs

Five backbones on FaceForensics++



Build Ensembles

Soft-voting baseline and stacking framework



Compare Meta-learners

Logistic Regression, Random Forest, XGBoost, MLP



Release Resources

Reproducible scripts & checkpoints





Real-World Applications

Social Media

Auto-flagging manipulated videos before upload

Digital Forensics

Validating video evidence in courts

Electoral Security

Detecting deepfake campaign ads

Finance (KYC)

Preventing face-swap fraud in remote onboarding

An abstract graphic on the left side of the slide featuring a network of glowing blue nodes connected by thin lines, with a bright blue beam of light passing through the center.

Dataset & Preprocessing



FaceForensics++ Dataset

~10,560 frames from "Deepfakes" subset



Frame Extraction

1 frame every 30 frames via OpenCV



Resize & Normalize

299×299 for Xception; 224×224 for others



Augmentation

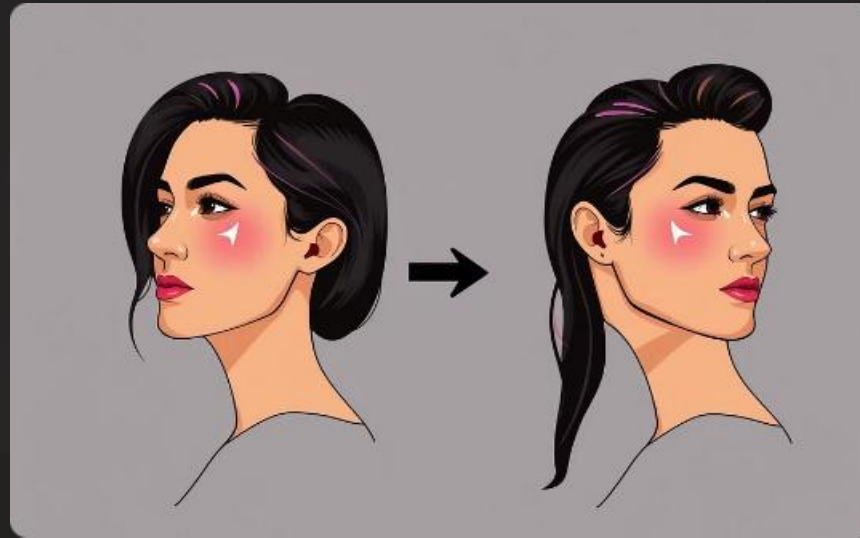
On-the-fly rotation, flips, shifts

Data Augmentation Techniques



Random Rotation

$\pm 20^\circ$ to simulate camera tilt



Horizontal Flip

Mirror face orientation



Noise & Jitter

Simulate compression artifacts

CNN Backbones Overview

Xception

Depthwise separable convolutions;
299×299 input

ResNet50

Residual connections; 50-layer
bottlenecks

EfficientNetB0

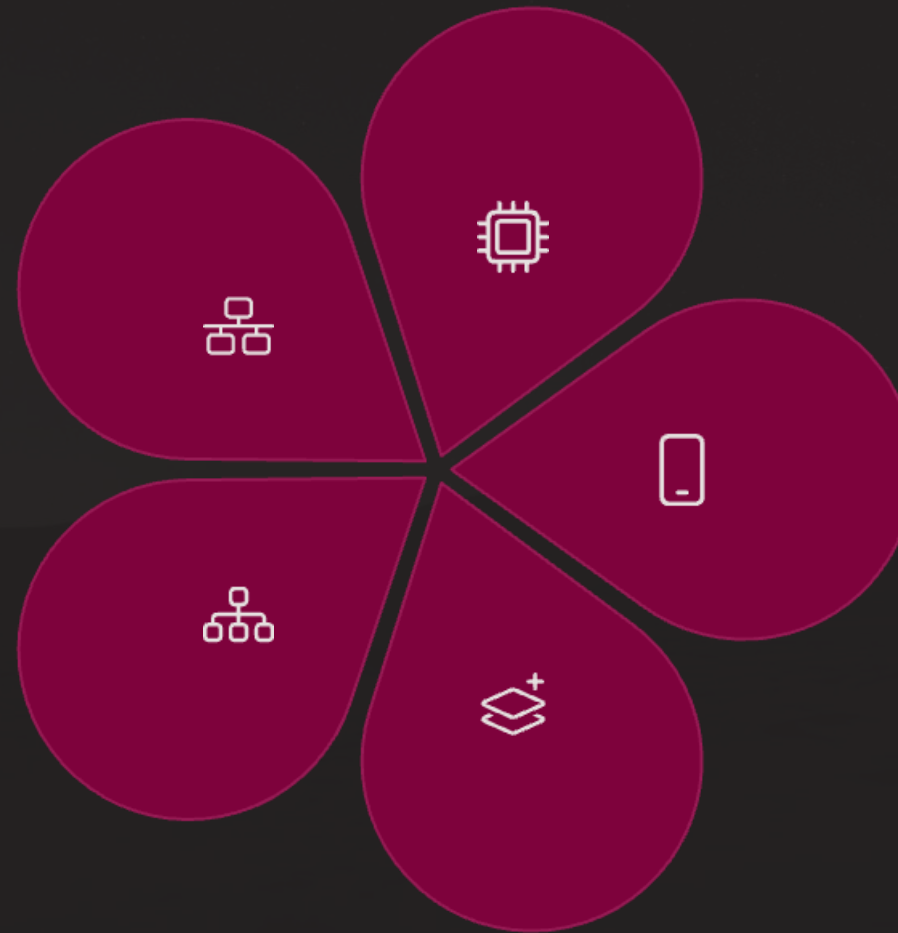
Compound scaling; 224×224 input

MobileNetV2

Inverted residuals; resource-
efficient

VGG19

Stack of 3×3 conv layers



Fine-Tuning Strategy



Layer Freezing

Freeze 70-85% of pretrained layers



Unfreeze Top Layers

Adapt high-level filters to forgery cues



Hyperparameter Tuning

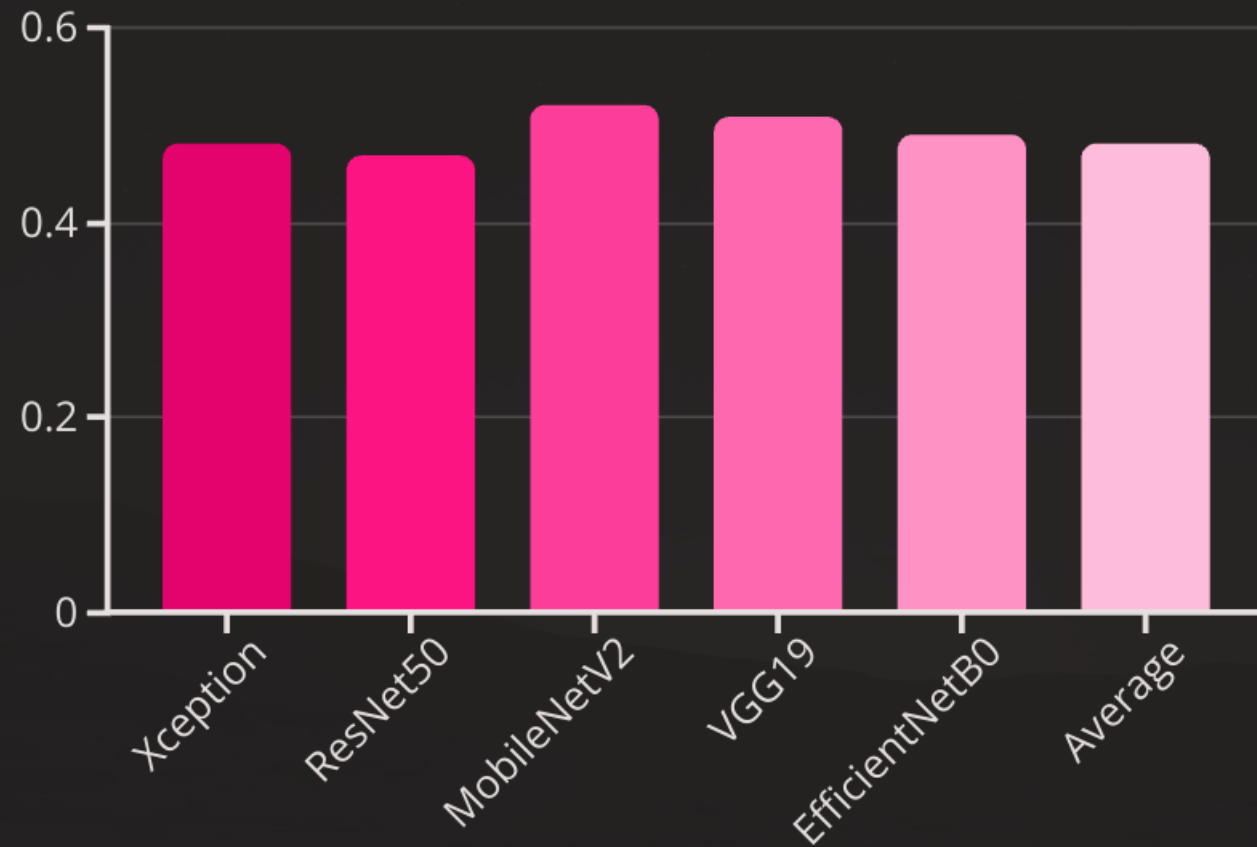
LR: 1×10^{-4} to 1×10^{-3} , Adam optimizer



Regularization

Dropout(0.5) in classifier head

Soft-Voting Ensemble



Formula: $p_{\text{soft}}(x) = (1/5) \sum_{i=1}^5 p_i(x)$

Decision: Fake if $p_{\text{soft}}(x) > 0.5$, else Real

Result: 47.0% accuracy, AUC=0.50 on hold-out set

Stacking Ensemble Concept



Meta-Features

Five CNN probabilities form vector $[p_1, \dots, p_5]$



Meta-Learner

Learns nonlinear fusion weights



Cross-Validation

5-fold stacking prevents label leakage



Key Insight

Stacking outperforms soft-voting

Meta-Feature Construction



Preprocess frame

Per-CNN requirements



Obtain probabilities

$p_i(x) \in [0,1]$ from sigmoid outputs



Concatenate

$x' = [p_1, p_2, p_3, p_4, p_5]$



Standardize

Zero mean, unit variance

Meta-Learners & Hyperparameters

Logistic Regression

L2 penalty, $C=1.0$, solver='lbfgs'

Random Forest

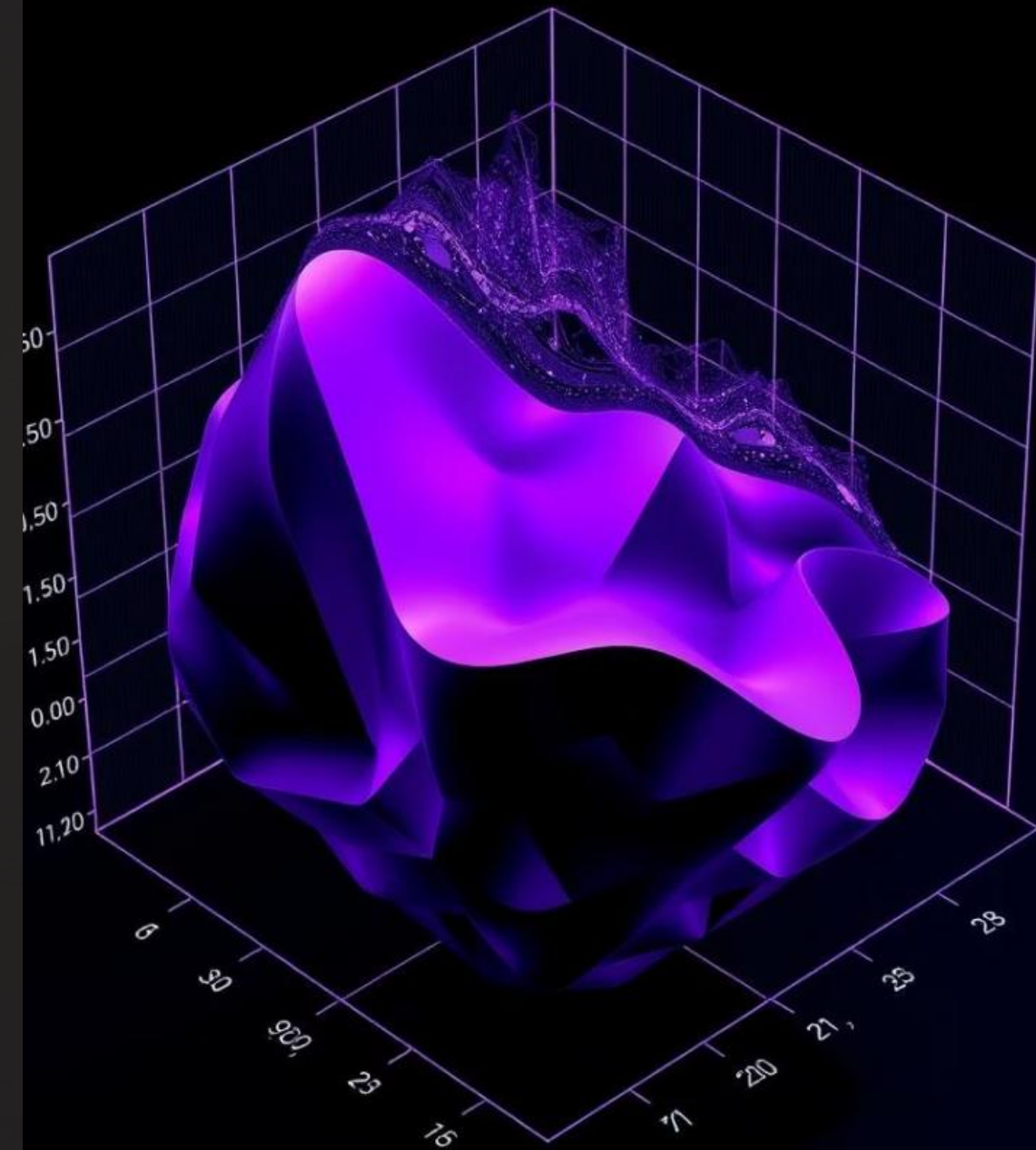
200 trees, max_depth=10, balanced sampling

XGBoost

300 rounds, max_depth=5, $\eta=0.1$

Neural Network

[64, 32] units, ReLU, dropout=0.5



Ensemble Performance Comparison

Meta-Model	Accuracy (%)	AUC
Soft Voting	47.0	0.50
Logistic Regression	57.6	0.59
Random Forest	65.5	0.65
XGBoost	64.9	0.72
Neural Network	66.4	0.67
RF (5-Fold CV)	67.1	0.68

ROC Curve & AUC Insights

Highest AUC

- XGBoost: 0.72
- Neural Network: 0.73
- RF: 0.68 (balanced)
- LR: 0.59 (limited)

Threshold Tuning

Choose based on FP/FN trade-off

Example

At FPR=0.1, XGBoost TPR $\approx$ 0.6 vs. LR TPR $\approx$ 0.45

Confusion Matrix & Calibration

RF & XGBoost

Balanced FP & FN (<120 each)



Neural Network

Overconfident predictions near 0/1



Confidence Histograms

RF bimodal peaks near extremes

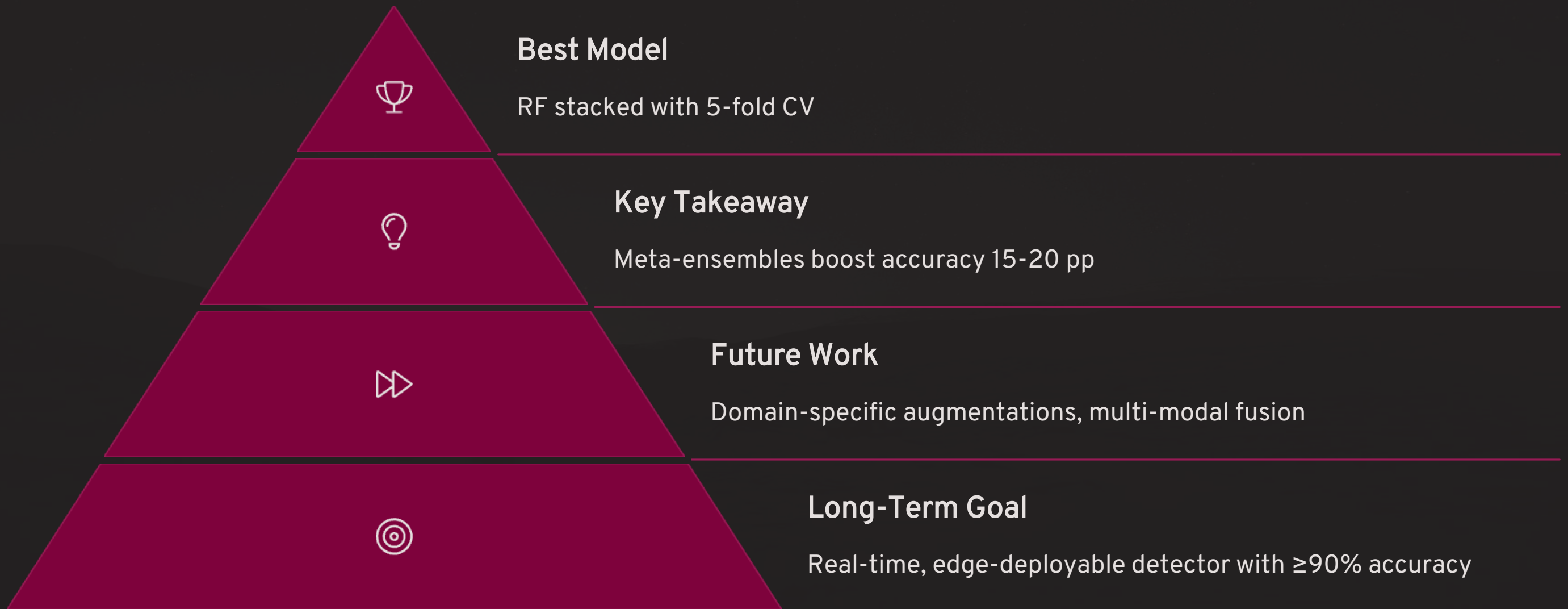


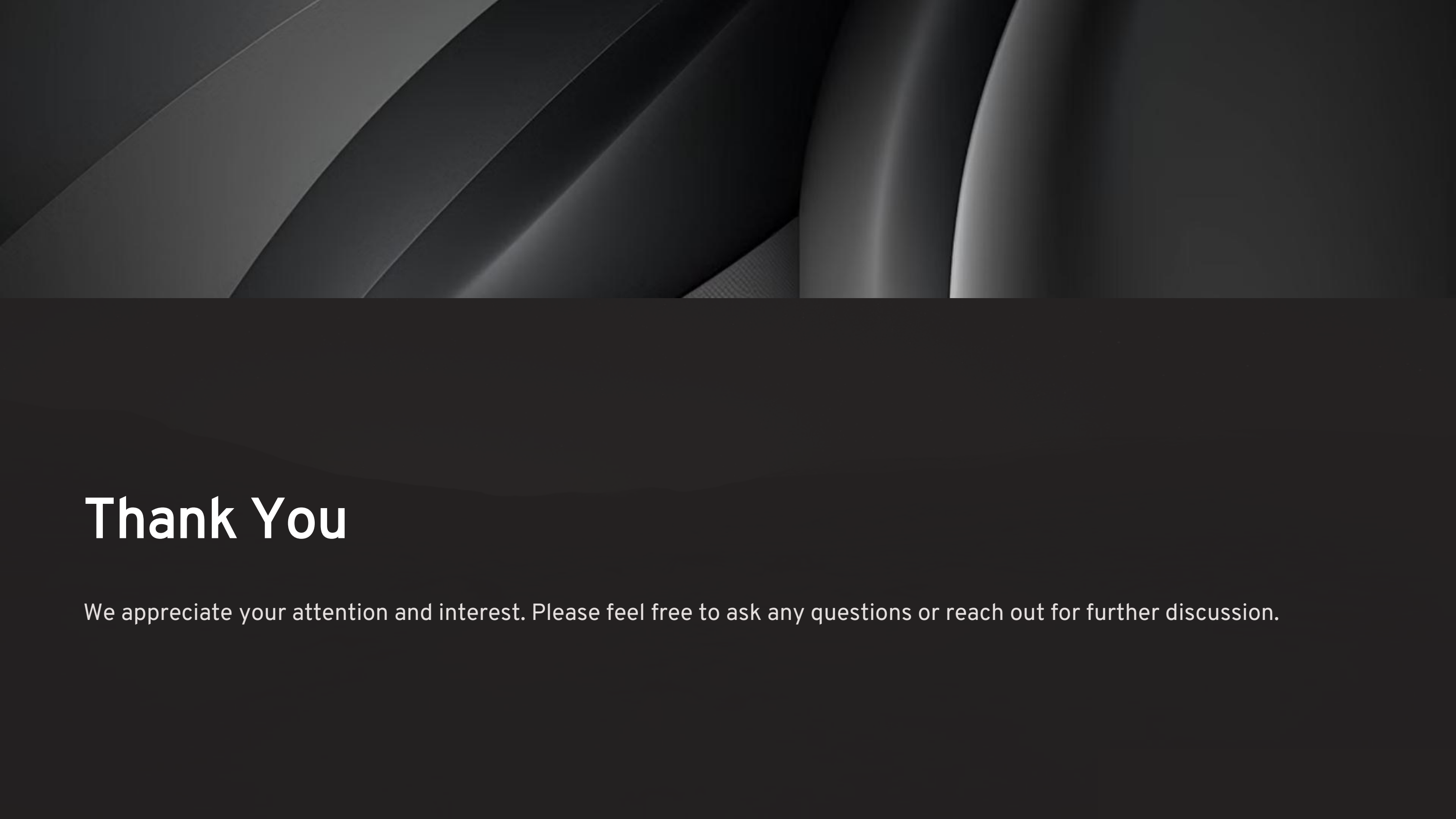
Logistic Regression

Ambiguous outputs → higher FN



Conclusion & Future Directions





Thank You

We appreciate your attention and interest. Please feel free to ask any questions or reach out for further discussion.